

ASME B31.3-2024

Table K-1

Allowable Stresses in Tension for Metals for Chapter IX — SI Units

(24)

Numbers in Parentheses Refer to Notes for Tables K-1 and K-1C ; Specifications Are ASTM Unless Otherwise Indicated									
Line No.	Nominal Composition	Product Form	Spec. No.	Type or Grade	UNS No.	P-No. (1)	Notes	Min. Tensile Strength, MPa	Min. Yield Strength MPa
1	Carbon steel	Smls. pipe	A53	B	K03005	1	(5) (7)	415	240
2	Carbon steel	Smls. pipe	A106	B	K03006	1	(5)	415	240
3	Carbon steel	Smls. pipe	A333	6	K03006	1	(5)	415	240
4	Carbon steel	Smls. tube	A334	6	K03006	1	(5)	415	240
5	Carbon steel	Smls. pipe	API 5L	B	...	1	(5) (6) (8) (10)	415	245
6	Carbon steel	Smls. tube	A210	A-1	K02707	1	(5)	415	255
7	Carbon steel	Smls. pipe	A106	C	K03501	1	(5)	485	275
8	Carbon steel	Smls. tube	A210	C	K03501	1	(5)	485	275
9	Carbon steel	Smls. pipe	API 5L	X42	...	1	(5) (6) (8) (10)	415	290
10	Carbon steel	Smls. pipe	API 5L	X46	...	1	(5) (6) (8) (10)	435	320
11	Carbon steel	Smls. pipe	API 5L	X52	...	1	(5) (6) (8) (10)	460	360
12	Carbon steel	Smls. pipe	API 5L	X56	...	1	(5) (6) (8) (10)	490	390
13	Carbon steel	Smls. pipe	API 5L	X60	...	1	(5) (6) (8) (10)	520	415
14	Carbon steel	Smls. pipe	API 5L	X65	...	1	(5) (6) (8) (10)	535	450
15	Carbon steel	Smls. pipe	API 5L	X70	...	1	(5) (6) (8) (10)	570	485
16	Carbon steel	Smls. pipe	API 5L	X80	...	1	(5) (6) (8) (10)	625	555
17	Carbon steel	Smls. ftgs.	A234	WPB	K03006	1	...	415	240
18	Carbon steel	Smls. ftgs.	A420	WPL6	K03006	1	...	415	240
19	Carbon steel	Forgings	A105	...	K03504	1	...	485	250
20	Carbon steel	Forgings	A350	LF2	K03011	1	...	485	250
21	Carbon steel	Smls. ftgs.	A234	WPC	K03501	1	...	485	275
22	Carbon steel	Forgings	A694	F42	K03014	1	...	415	290
23	Carbon steel	Forgings	A694	F46	K03014	1	...	415	315
24	Carbon steel	Forgings	A694	F48	K03014	1	...	425	330
25	Carbon steel	Forgings	A694	F50	K03014	1	...	440	345
26	Carbon steel	Forgings	A694	F52	K03014	1	...	455	360
27	Carbon steel	Forgings	A694	F56	K03014	1	...	470	385
28	Carbon steel	Forgings	A694	F60	K03014	1	...	515	415
29	Carbon steel	Forgings	A694	F65	K03014	1	...	530	450
30	Carbon steel	Forgings	A694	F70	K03014	1	...	565	485

ASME B31.3-2024

Table K-1
Allowable Stresses in Tension for Metals for Chapter IX — SI Units

Numbers in Parentheses Refer to Notes for Tables K-1 and K-1C ; Specifications Are ASTM Unless Otherwise Indicated											
Line No.	Allowable Stress, S, MPa, at Metal Temperature, °C, Not Exceeding [Notes (2)–(4)]										
	Max. Temp., °C	Min. Temp. to 40	65	100	150	200	250	300	325	350	375
1	371	212	199	193	188	182	174	165	161	155	151
2	371	212	199	193	188	182	174	165	161	155	151
3	371	212	199	193	188	182	174	165	161	155	151
4	371	212	199	193	188	182	174	165	161	155	151
5	371	212	199	193	188	182	174	165	161	155	151
6	371	223	211	205	198	192	184	176	169	164	160
7	371	242	227	220	214	208	199	190	183	177	172
8	371	242	227	220	214	208	199	190	183	177	172
9	260	236	229	223	207	194	182	174	...	...	...
10	260	251	243	237	222	208	196	190	...	...	...
11	260	274	265	259	247	231	218	207	...	...	...
12	260	292	283	276	264	247	233	224	...	...	...
13	260	312	302	295	282	264	249	238	...	...	...
14	260	329	318	310	300	282	265	252	...	...	...
15	204	353	341	332	321	302	284	...	...	...	...
16	204	392	384	372	358	336	315	...	...	...	...
17	371	212	199	193	188	182	174	165	161	155	151
18	371	212	199	193	188	182	174	165	161	155	151
19	371	218	205	199	192	187	179	170	165	161	155
20	371	218	205	199	192	187	179	170	165	161	155
21	371	242	227	220	214	208	199	190	183	177	172
22	260	236	229	223	207	194	182	174	...	...	...
23	260	244	236	231	218	205	193	184	...	...	...
24	260	252	244	238	227	212	201	191	...	...	...
25	260	262	254	248	236	222	209	200	...	...	...
26	260	272	264	257	246	230	217	208	...	...	...
27	260	286	276	270	260	243	229	219	...	...	...
28	260	311	301	293	281	264	249	238	...	...	...
29	260	328	317	309	299	280	264	253	...	...	...
30	204	351	339	330	320	301	283	...	...	...	...

ASME B31.3-2024

Table K-1
Allowable Stresses in Tension for Metals for Chapter IX — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables K-1 and K-1C ; Specifications Are ASTM Unless Otherwise Indicated									
Line No.	Nominal Composition	Product Form	Spec. No.	Type or Grade	UNS No.	P-No. (1)	Notes	Min. Tensile Strength, MPa	Min. Yield Strength, MPa
1	C- $\frac{1}{2}$ Mo	Smls. pipe	A335	P1	K11522	3	(5)	380	205
2	1Cr- $\frac{1}{2}$ Mo	Smls. pipe	A335	P12	K11562	4	(5)	415	220
3	1 $\frac{1}{4}$ Cr- $\frac{1}{2}$ Mo-Si	Smls. pipe	A335	P11	K11597	4	(5)	415	205
4	2 $\frac{1}{4}$ Cr-1Mo	Smls. pipe	A335	P22	K21590	5A	(5)	415	205
5	5Cr- $\frac{1}{2}$ Mo	Smls. pipe	A335	P5	K41545	5B	(5)	415	205
6	3 $\frac{1}{2}$ Ni	Smls. pipe	A333	3	K31918	9B	(5)	450	240
7	3 $\frac{1}{2}$ Ni	Smls. tube	A334	3	K31918	9B	(5)	450	240
8	9Ni	Smls. pipe	A333	8	K81340	11A	(5)	690	515
9	9Ni	Smls. tube	A334	8	K81340	11A	(5)	690	515
10	C- $\frac{1}{2}$ Mo	Forgings	A182	F1	K12822	3	...	485	275
11	1Cr- $\frac{1}{2}$ Mo	Forgings	A182	F12, Cl. 2	K11564	4	...	485	275
12	1 $\frac{1}{4}$ Cr- $\frac{1}{2}$ Mo-Si	Forgings	A182	F11, Cl. 2	K11572	4	...	485	275
13	2 $\frac{1}{4}$ Cr-1Mo	Forgings	A182	F22, Cl. 3	K21590	5A	...	515	310
14	5Cr- $\frac{1}{2}$ Mo	Forgings	A182	F5	K41545	5B	...	485	275
15	2Ni-1 $\frac{1}{2}$ Cr- $\frac{1}{4}$ Mo-V	Forgings	A723	1, Cl. 1	K23550	...	(11)	795	690
16	2Ni-1 $\frac{1}{2}$ Cr- $\frac{1}{4}$ Mo-V	Forgings	A723	1, Cl. 2	K23550	...	(11)	930	825
17	2Ni-1 $\frac{1}{2}$ Cr- $\frac{1}{4}$ Mo-V	Forgings	A723	1, Cl. 3	K23550	...	(11)	1070	965
18	2 $\frac{3}{4}$ Ni-1 $\frac{1}{2}$ Cr- $\frac{1}{2}$ Mo-V	Forgings	A723	2, Cl. 1	K23550	...	(11)	795	690
19	2 $\frac{3}{4}$ Ni-1 $\frac{1}{2}$ Cr- $\frac{1}{2}$ Mo-V	Forgings	A723	2, Cl. 2	K23550	...	(11)	930	825
20	2 $\frac{3}{4}$ Ni-1 $\frac{1}{2}$ Cr- $\frac{1}{2}$ Mo-V	Forgings	A723	2, Cl. 3	K23550	...	(11)	1070	965
21	3 $\frac{1}{2}$ Ni	Smls. ftgs.	A420	WPL3	K31918	9B	...	450	240
22	3 $\frac{1}{2}$ Ni	Forgings	A350	LF3	K32025	9B	...	485	260
23	3 $\frac{1}{2}$ Ni-1 $\frac{3}{4}$ Cr- $\frac{1}{2}$ Mo-V	Forgings	A508	4N, Cl. 2	K22375	11B	...	795	690
24	4Ni-1 $\frac{1}{2}$ Cr- $\frac{1}{2}$ Mo-V	Forgings	A723	3, Cl. 1	K23550	...	(11)	795	690
25	4Ni-1 $\frac{1}{2}$ Cr- $\frac{1}{2}$ Mo-V	Forgings	A723	3, Cl. 2	K23550	...	(11)	930	825
26	4Ni-1 $\frac{1}{2}$ Cr- $\frac{1}{2}$ Mo-V	Forgings	A723	3, Cl. 3	K23550	...	(11)	1070	965
27	9Ni	Smls. ftgs.	A420	WPL8	K81340	11A	...	690	515

ASME B31.3-2024

Table K-1
Allowable Stresses in Tension for Metals for Chapter IX — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables K-1 and K-1C ; Specifications Are ASTM Unless Otherwise Indicated															
Line No.	Max. Temp., °C	Allowable Stress, S, MPa, at Metal Temperature, °C, Not Exceeding [Notes (2)–(4)]													
		Min. Temp. to 40	65	100	125	150	175	200	225	250	275	300	325	350	375
1	371	182	175	169	167	164	162	158	156	154	152	149	147	145	142
2	371	194	182	174	169	164	161	158	155	153	151	149	147	145	143
3	371	182	173	167	162	159	156	154	151	148	146	144	141	139	136
4	371	182	173	169	167	164	163	162	162	162	162	162	162	162	162
5	371	182	169	163	160	158	157	156	155	155	154	154	152	150	147
6	371	212	199	193	190	187	184	182	178	174	169	163	156	149	141
7	371	212	199	193	190	187	184	182	178	174	169	163	156	149	141
8	149	402	397	395	390	382	...	...	...	...	...	...	...	...	...
9	149	402	397	395	390	382	...	...	...	...	...	...	...	...	...
10	371	242	233	226	222	219	215	212	208	205	203	200	197	193	190
11	371	242	227	218	211	205	201	198	194	191	189	186	183	182	179
12	371	242	230	222	217	212	208	205	201	198	195	192	189	185	183
13	371	272	257	248	242	238	234	231	228	226	225	222	220	218	215
14	371	242	226	218	213	211	209	208	208	207	205	205	203	200	197
15	371	494	489	484	481	479	477	476	474	473	472	470	467	459	450
16	371	586	579	574	571	568	566	564	563	561	559	557	554	544	533
17	371	678	671	664	661	658	655	653	651	649	647	645	642	631	616
18	371	494	489	484	481	479	477	476	474	473	472	470	467	459	450
19	371	586	579	574	571	568	566	564	563	561	559	557	554	544	533
20	371	678	671	664	661	658	655	653	651	649	647	645	642	631	616
21	343	212	199	193	190	187	184	182	178	174	169	163	156	149	...
22	343	227	214	206	204	201	198	194	190	186	181	175	168	160	...
23	343	494	486	480	476	474	470	467	465	463	460	456	453	447	...
24	371	494	489	484	481	479	477	476	474	473	472	470	467	459	450
25	371	586	579	574	571	568	566	564	563	561	559	557	554	544	533
26	371	678	671	664	661	658	655	653	651	649	647	645	642	631	616
27	149	402	397	395	390	382	...	...	...	...	...	...	...	...	...

ASME B31.3-2024

Table K-1
Allowable Stresses in Tension for Metals for Chapter IX — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables K-1 and K-1C; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type or Grade	UNS No.	Thickness Range, mm	P-No. (1)	Notes
1	16Cr-12Ni-2Mo	Smls. & wld. pipe	A312	TP316L	S31603	...	8	(5) (13)
2	16Cr-12Ni-2Mo	Wld. pipe	A358	316L, Cl. 1 & 3	S31603	...	8	(5) (13) (14)
3	16Cr-12Ni-2Mo-N	Smls. & wld. pipe	A312	TP316LN	S31653	...	8	(5) (13)
4	16Cr-12Ni-2Mo-N	Wld. pipe	A358	316LN, Cl. 1 & 3	S31653	...	8	(5) (13) (14)
5	18Cr-8Ni	Smls. & wld. pipe	A312	TP304L	S30403	...	8	(5) (13)
6	18Cr-8Ni	Wld. pipe	A358	304L, Cl. 1 & 3	S30403	...	8	(5) (13) (14)
7	18Cr-8Ni-N	Smls. & wld. pipe	A312	TP304LN	S30453	...	8	(5) (13)
8	18Cr-8Ni-N	Wld. pipe	A358	304LN, Cl. 1 & 3	S30453	...	8	(5) (13) (14)
9	18Cr-10Ni-Ti	Smls. pipe	A312	TP321	S32100	>9.50	8	(5)
10	18Cr-10Ni-Ti	Smls. pipe	A312	TP321	S32100	≤9.50	8	(5)
11	18Cr-10Ni-Ti	Wld. pipe	A312	TP321	S32100	...	8	(5) (13)
12	18Cr-10Ni-Ti	Wld. pipe	A358	321, Cl. 1 & 3	S32100	...	8	(5) (13) (14)
13	18Cr-8Ni	Smls. & wld. pipe	A312	TP304	S30400	...	8	(5) (13) (15)
14	18Cr-8Ni	Wld. pipe	A358	304, Cl. 1 & 3	S30400	...	8	(5) (13) (14) (15)
15	16Cr-12Ni-2Mo	Smls. & wld. pipe	A312	TP316	S31600	...	8	(5) (13) (15)
16	16Cr-12Ni-2Mo	Wld. pipe	A358	316, Cl. 1 & 3	S31600	...	8	(5) (13) (14) (15)
17	18Cr-13Ni-3Mo	Smls. & wld. pipe	A312	TP317	S31700	...	8	(5) (13) (15)
18	18Cr-10Ni-Cb	Smls. & wld. pipe	A312	TP347	S34700	...	8	(5) (13)
19	18Cr-10Ni-Cb	Wld. pipe	A358	347, Cl. 1 & 3	S34700	...	8	(5) (13) (14)
20	18Cr-8Ni-N	Smls. & wld. pipe	A312	TP304N	S30451	...	8	(5) (13) (15)
21	18Cr-8Ni-N	Wld. pipe	A358	304N, Cl. 1 & 3	S30451	...	8	(5) (13) (14) (15)
22	16Cr-12Ni-2Mo-N	Smls. & wld. pipe	A312	TP316N	S31651	...	8	(5) (13) (15)
23	16Cr-12Ni-2Mo-N	Wld. pipe	A358	316N, Cl. 1 & 3	S31651	...	8	(5) (13) (14) (15)
24	22Cr-5Ni-3Mo-N	Smls. & wld. tube	A789	...	S31803	...	10H	(5) (9) (13)
25	22Cr-5Ni-3Mo-N	Smls. & wld. pipe	A790	...	S31803	...	10H	(5) (9) (13)
26	22Cr-5Ni-3Mo-N	Smls. & wld. pipe	A790	2205	S32205	...	10H	(5) (9) (13)
27	22Cr-5Ni-3Mo-N	Wld. pipe	A928	Cl. 1, 3, & 4	S31803	...	10H	(5) (9) (13)
28	22Cr-5Ni-3Mo-N	Wld. pipe	A928	2205, Cl. 1, 3, & 4	S32205	...	10H	(5) (9) (13)
29	22Cr-5Ni-3Mo-N	Smls. & wld. tube	A789	...	S32205	...	10H	(5) (9) (13)

ASME B31.3-2024

Table K-1
Allowable Stresses in Tension for Metals for Chapter IX — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables K-1 and K-1C ; Specifications Are ASTM Unless Otherwise Indicated																	
Line No.	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Allowable Stress, S, MPa, at Metal Temperature, °C, Not Exceeding [Notes (2)–(4)]													
				Min. Temp. to 40	65	100	125	150	175	200	225	250	275	300	325	350	375
1	485	170	371	151	138	127	120	115	110	106	104	100	97	96	94	92	90
2	485	170	371	151	138	127	120	115	110	106	104	100	97	96	94	92	90
3	515	205	371	182	164	153	145	139	133	128	124	119	116	112	110	107	104
4	515	205	371	182	164	153	145	139	133	128	124	119	116	112	110	107	104
5	485	170	371	151	138	128	121	116	111	106	103	100	97	95	93	91	90
6	485	170	371	151	138	128	121	116	111	106	103	100	97	95	93	91	90
7	515	205	371	182	162	149	141	135	130	126	122	119	116	113	111	108	106
8	515	205	371	182	162	149	141	135	130	126	122	119	116	113	111	108	106
9	485	170	371	151	142	135	129	126	121	117	113	110	106	104	102	99	97
10	515	205	371	182	170	162	155	150	145	140	135	132	128	125	122	119	117
11	515	205	371	182	170	162	155	150	145	140	135	132	128	125	122	119	117
12	515	205	371	182	170	162	155	150	145	140	135	132	128	125	122	119	117
13	515	205	371	182	162	149	141	135	130	126	122	119	116	113	111	108	106
14	515	205	371	182	162	149	141	135	130	126	122	119	116	113	111	108	106
15	515	205	371	182	166	154	147	141	135	130	126	122	119	116	113	111	110
16	515	205	371	182	166	154	147	141	135	130	126	122	119	116	113	111	110
17	515	205	371	182	166	154	147	141	135	130	126	122	119	116	113	111	110
18	515	205	371	182	173	166	160	155	151	146	142	138	135	132	129	127	125
19	515	205	371	182	173	166	160	155	151	146	142	138	135	132	129	127	125
20	550	240	371	212	187	170	160	151	144	138	133	128	125	122	119	118	115
21	550	240	371	212	187	170	160	151	144	138	133	128	125	122	119	118	115
22	550	240	371	212	197	186	179	172	166	161	155	151	147	144	140	137	134
23	550	240	371	212	197	186	179	172	166	161	155	151	147	144	140	137	134
24	620	450	260	356	346	338	330	322	314	308	303	299	295	...	...	...	...
25	620	450	260	356	346	338	330	322	314	308	303	299	295	...	...	...	...
26	655	450	260	368	358	347	334	325	317	311	306	302	298	...	...	...	...
27	620	450	260	356	346	338	330	322	314	308	303	299	295	...	...	...	...
28	655	450	260	368	358	347	334	325	317	311	306	302	298	...	...	...	...
29	655	485	260	379	368	360	351	343	336	330	325	319	315	...	...	...	...

ASME B31.3-2024

Table K-1
Allowable Stresses in Tension for Metals for Chapter IX — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables K-1 and K-1C; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type or Grade	UNS No.	Thickness Range, mm	P-No. (1)	Notes
30	25Cr-7Ni-4Mo-N	Smls. & wld. tube	A789	...	S32750	...	10H	(5) (9) (13)
31	25Cr-7Ni-4Mo-N	Smls. & wld. pipe	A790	2507	S32750	...	10H	(5) (9) (13)
32	25Cr-7Ni-4Mo-N	Wld. pipe	A928	2507, Cl. 1, 3, & 4	S32750	...	10H	(5) (9) (13)
33	16Cr-12Ni-2Mo	Forgings	A182	F316L	S31603	...	8	(16)
34	16Cr-12Ni-2Mo	Smls. & wld. ftgs.	A403	WP316L, Cl. S & WX	S31603	...	8	(13)
35	16Cr-12Ni-2Mo-N	Forgings	A182	F316LN	S31653	...	8	(17)
36	16Cr-12Ni-2Mo-N	Smls. & wld. ftgs.	A403	WP316LN, Cl. S & WX	S31653	...	8	(13)
37	18Cr-8Ni	Forgings	A182	F304L	S30403	...	8	(16)
38	18Cr-8Ni	Smls. & wld. ftgs.	A403	WP304L, Cl. S & WX	S30403	...	8	(13)
39	18Cr-8Ni-N	Forgings	A182	F304LN	S30453	...	8	(17)
40	18Cr-8Ni-N	Smls. & wld. ftgs.	A403	WP304LN, Cl. S & WX	S30453	...	8	(13)
41	18Cr-10Ni-Ti	Forgings	A182	F321	S32100	...	8	(17)
42	18Cr-10Ni-Ti	Smls. & wld. ftgs.	A403	WP321, Cl. S & WX	S32100	...	8	(13)
43	18Cr-8Ni	Forgings	A182	F304	S30400	...	8	(15) (17)
44	18Cr-8Ni	Smls. & wld. ftgs.	A403	WP304, Cl. S & WX	S30400	...	8	(13) (15)
45	16Cr-12Ni-2Mo	Forgings	A182	F316	S31600	...	8	(15) (17)
46	16Cr-12Ni-2Mo	Smls. & wld. ftgs.	A403	WP316, Cl. S & WX	S31600	...	8	(13) (15)
47	18Cr-13Ni-3Mo	Smls. & wld. ftgs.	A403	WP317, Cl. S & WX	S31700	...	8	(13) (15)
48	18Cr-10Ni-Cb	Forgings	A182	F347	S34700	...	8	(17)
49	18Cr-10Ni-Cb	Smls. & wld. ftgs.	A403	WP347, Cl. S & WX	S34700	...	8	(13)
50	18Cr-8Ni-N	Forgings	A182	F304N	S30451	...	8	(15)
51	18Cr-8Ni-N	Smls. & wld. ftgs.	A403	WP304N, Cl. S & WX	S30451	...	8	(13) (15)
52	16Cr-12Ni-2Mo-N	Forgings	A182	F316N	S31651	...	8	(15)
53	16Cr-12Ni-2Mo-N	Smls. & wld. ftgs.	A403	WP316N, Cl. S & WX	S31651	...	8	(13) (15)
54	22Cr-5Ni-3Mo-N	Forgings	A182	F51	S31803	...	10H	(9)
55	22Cr-5Ni-3Mo-N	Smls. & wld. ftgs.	A815	WP31803, Cl. WP-S & WP-WX	S31803	...	10H	(9) (13)
56	22Cr-5Ni-3Mo-N	Forgings	A182	F60	S32205	...	10H	(9)
57	22Cr-5Ni-3Mo-N	Smls. & wld. ftgs.	A815	WP32205, Cl. WP-S & WP-WX	S32205	...	10H	(9) (13)
58	25Cr-7Ni-4Mo-N	Forgings	A182	F53	S32750	...	10H	(9)
59	25Cr-7Ni-4Mo-N	Smls. & wld. ftgs.	A815	WP32750, Cl. WP-S & WP-WX	S32750	...	10H	(9) (13)

ASME B31.3-2024

Table K-1
Allowable Stresses in Tension for Metals for Chapter IX — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables K-1 and K-1C; Specifications Are ASTM Unless Otherwise Indicated																	
Line No.	Allowable Stress, S, MPa, at Metal Temperature, °C, Not Exceeding [Notes (2)–(4)]																
	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Min. Temp. to 40	65	100	125	150	175	200	225	250	275	300	325	350	375
30	800	550	260	451	436	422	406	391	379	369	361	355	350	...	...	...	...
31	800	550	260	451	436	422	406	391	379	369	361	355	350	...	...	...	...
32	800	550	260	451	436	422	406	391	379	369	361	355	350	...	...	...	...
33	485	170	371	151	138	127	120	115	110	106	104	100	97	96	94	92	90
34	485	170	371	151	138	127	120	115	110	106	104	100	97	96	94	92	90
35	515	205	371	182	164	153	145	139	133	128	124	119	116	112	110	107	104
36	515	205	371	182	164	153	145	139	133	128	124	119	116	112	110	107	104
37	485	170	371	151	138	128	121	116	111	106	103	100	97	95	93	91	90
38	485	170	371	151	138	128	121	116	111	106	103	100	97	95	93	91	90
39	515	205	371	182	162	149	141	135	130	126	122	119	116	113	111	108	106
40	515	205	371	182	162	149	141	135	130	126	122	119	116	113	111	108	106
41	515	205	371	182	170	162	155	150	145	140	135	132	128	125	122	119	117
42	515	205	371	182	170	162	155	150	145	140	135	132	128	125	122	119	117
43	515	205	371	182	162	149	141	135	130	126	122	119	116	113	111	108	106
44	515	205	371	182	162	149	141	135	130	126	122	119	116	113	111	108	106
45	515	205	371	182	166	154	147	141	135	130	126	122	119	116	113	111	110
46	515	205	371	182	166	154	147	141	135	130	126	122	119	116	113	111	110
47	515	205	371	182	166	154	147	141	135	130	126	122	119	116	113	111	110
48	515	205	371	182	173	166	160	155	151	146	142	138	135	132	129	127	125
49	515	205	371	182	173	166	160	155	151	146	142	138	135	132	129	127	125
50	550	240	371	212	187	170	160	151	144	138	133	128	125	122	119	118	115
51	550	240	371	212	187	170	160	151	144	138	133	128	125	122	119	118	115
52	550	240	371	212	197	186	179	172	166	161	155	151	147	144	140	137	134
53	550	240	371	212	197	186	179	172	166	161	155	151	147	144	140	137	134
54	620	450	260	356	346	338	330	322	314	308	303	299	295	...	...	...	...
55	620	450	260	356	346	338	330	322	314	308	303	299	295	...	...	...	...
56	655	450	260	368	358	347	334	325	317	311	306	302	298	...	...	...	...
57	655	450	260	368	358	347	334	325	317	311	306	302	298	...	...	...	...
58	800	550	260	451	436	422	406	391	379	369	361	355	350	...	...	...	...
59	800	550	260	451	436	422	406	391	379	369	361	355	350	...	...	...	...

ASME B31.3-2024

Table K-1
Allowable Stresses in Tension for Metals for Chapter IX — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables K-1 and K-1C ; Specifications Are ASTM Unless Otherwise Indicated										
Line No.	Nominal Composition	Product Form	Spec. No.	UNS No.	Condition	Size or Thickness Range, mm	P-No. (1)	Notes	Min. Tensile Strength, MPa	Min. Yield Strength, MPa
1	67Ni-30Cu	Smls. pipe & tube	B165	N04400	Annealed	>127 O.D.	42	(5)	485	170
2	72Ni-15Cu-8Fe	Smls. pipe & tube	B167	N06600	H.W.	>127 O.D.	43	(5)	515	170
3	72Ni-15Cu-8Fe	Smls. pipe & tube	B167	N06600	H.W. ann.	>127 O.D.	43	(5)	515	170
4	67Ni-30Cu	Smls. pipe & tube	B165	N04400	Annealed	≤127 O.D.	42	(5)	485	195
5	72Ni-15Cu-8Fe	Smls. pipe & tube	B167	N06600	H.W.	≤127 O.D.	43	(5)	550	205
6	72Ni-15Cu-8Fe	Smls. pipe & tube	B167	N06600	H.W. ann.	≤127 O.D.	43	(5)	550	205
7	72Ni-15Cu-8Fe	Smls. pipe & tube	B167	N06600	C.W. ann.	>127 O.D.	43	(5)	550	205
8	72Ni-15Cu-8Fe	Smls. pipe & tube	B167	N06600	C.W. ann.	≤127 O.D.	43	(5)	550	240
9	54Ni-16Mo-15Cr	Smls. pipe & tube	B622	N10276	...	All	43	(5)	690	285
10	67Ni-30Cu	Smls. pipe & tube	B165	N04400	Str. rel.	All	42	(5)	585	380
11	67Ni-30Cu	Smls. & wld. ftgs.	B366	N04400	...	All	42	(12) (13)	485	170
12	67Ni-30Cu	Forgings	B564	N04400	Annealed	All	42	...	485	170
13	72Ni-15Cu-8Fe	Smls. & wld. ftgs.	B366	N06600	...	All	43	(12) (13)	515	170
14	72Ni-15Cu-8Fe	Forgings	B564	N06600	Annealed	All	43	...	550	240
15	54Ni-16Mo-15Cr	Smls. & wld. ftgs.	B366	N10276	...	All	43	(13)	690	285
16	54Ni-16Mo-15Cr	Forgings	B564	N10276	Annealed	All	43	...	690	285
17	67Ni-30Cu	Rod & bar	B164	N04400	Annealed	All	42	...	485	170
18	72Ni-15Cu-8Fe	Rod & bar	B166	N06600	C.W. ann. & H.W. ann.	All	43	...	550	240
19	72Ni-15Cu-8Fe	Rod & bar	B166	N06600	H.W., A.W.	Sq., rec. & hex. All	43	...	585	240
20	72Ni-15Cu-8Fe	Rod & bar	B166	N06600	H.W., A.W.	>76.2 rd.	43	...	585	240
21	72Ni-15Cu-8Fe	Rod & bar	B166	N06600	H.W., A.W.	>12.7 rd., ≤76.2 rd.	43	...	620	275
22	54Ni-16Mo-15Cr	Rod	B574	N10276	...	All	43	...	690	285
23	72Ni-15Cu-8Fe	Rod & bar	B166	N06600	H.W., A.W.	≥6.5 rd., ≤12.7 rd.	43	...	655	310

ASME B31.3-2024

Table K-1
Allowable Stresses in Tension for Metals for Chapter IX — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables K-1 and K-1C ; Specifications Are ASTM Unless Otherwise Indicated															
Line No.	Max. Temp., °C	Allowable Stress, S, MPa, at Metal Temperature, °C, Not Exceeding [Notes (2)–(4)]													
		Min. Temp. to 40	65	100	125	150	175	200	225	250	275	300	325	350	375
1	371	151	140	131	126	124	121	119	119	119	119	119	119	119	119
2	371	151	147	143	140	138	135	133	130	128	126	124	122	120	119
3	371	151	147	143	140	138	135	133	130	128	126	124	122	120	119
4	371	169	156	147	142	138	136	134	133	133	133	133	133	133	133
5	371	182	176	172	169	166	162	159	156	154	151	148	147	144	142
6	371	182	176	172	169	166	162	159	156	154	151	148	147	144	142
7	371	182	176	172	169	166	162	159	156	154	151	148	147	144	142
8	371	212	198	193	190	189	187	186	185	183	183	182	181	179	177
9	371	248	235	224	216	208	201	195	189	183	177	172	168	164	161
10	260	322	317	312	308	305	303	301	300	298	296	...	...	...	...
11	371	151	140	131	126	124	121	119	119	119	119	119	119	119	119
12	371	151	140	131	126	124	121	119	119	119	119	119	119	119	119
13	371	151	147	143	140	138	135	133	130	128	126	124	122	120	119
14	371	212	198	193	190	189	187	186	185	183	183	182	181	179	177
15	371	248	235	224	216	208	201	195	189	183	177	172	168	164	161
16	371	248	235	224	216	208	201	195	189	183	177	172	168	164	161
17	371	151	140	131	126	124	121	119	119	119	119	119	119	119	119
18	371	212	198	193	190	189	187	186	185	183	183	182	181	179	177
19	371	212	205	200	198	196	194	193	193	193	193	193	192	191	190
20	371	212	205	200	198	196	194	193	193	193	193	193	192	191	190
21	371	242	235	229	226	224	222	220	220	220	220	220	219	219	217
22	371	248	235	224	216	208	201	195	189	183	177	172	168	164	161
23	371	272	265	258	255	252	250	248	248	248	248	248	247	246	244

ASME B31.3-2024

Table K-1
Allowable Stresses in Tension for Metals for Chapter IX — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables K-1 and K-1C ; Specifications Are ASTM Unless Otherwise Indicated									
Line No.	Nominal Composition	Product Form	Spec. No.	Grade	UNS No.	P-No. (1)	Notes	Min. Tensile Strength, MPa	Min. Yield Strength, MPa
1	Ti	Smls. pipe	B861	2	R50400	51	(5)	345	275
2	Ti	Smls. tube	B338	2	R50400	51	(5)	345	275
3	Ti-Pd	Smls. pipe	B861	7	R52400	51	(5)	345	275
4	Ti-Pd	Smls. tube	B338	7	R52400	51	(5)	345	275
5	Ti	Smls. pipe	B861	3	R50550	52	(5)	450	380
6	Ti	Smls. tube	B338	3	R50550	52	(5)	450	380
7	Ti	Smls. ftgs.	B363	WPT2	R50400	51	...	345	275
8	Ti	Forgings	B381	F-2	R50400	51	...	345	275
9	Ti-Pd	Forgings	B381	F-7	R52400	51	...	345	275
10	Ti	Smls. ftgs.	B363	WPT3	R50550	52	...	450	380
11	Ti	Forgings	B381	F-3	R50550	52	...	450	380

Table K-1
Allowable Stresses in Tension for Metals for Chapter IX — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables K-1 and K-1C; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Max. Temp., °C	Allowable Stress, S, MPa, at Metal Temperature, °C, Not Exceeding [Notes (2)–(4)]											
		Min. Temp. to 40	65	100	125	150	175	200	225	250	275	300	325
1	316	207	190	168	154	141	129	116	102	91	78	69	64
2	316	207	190	168	154	141	129	116	102	91	78	69	64
3	316	207	190	168	154	141	129	116	102	91	78	69	64
4	316	207	190	168	154	141	129	116	102	91	78	69	64
5	316	276	250	223	203	184	167	152	138	125	114	104	92
6	316	276	250	223	203	184	167	152	138	125	114	104	92
7	316	207	187	168	155	141	129	116	102	91	78	69	64
8	316	207	190	168	154	141	129	116	102	91	78	69	64
9	316	207	190	168	154	141	129	116	102	91	78	69	64
10	316	276	250	223	203	184	167	152	138	125	114	104	92
11	316	276	250	223	203	184	167	152	138	125	114	104	92